Information Loss in Spatial Statistics Analysis: Entropy, Scale, and Uncertainty

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Abstract: Spatial data often contain several layers of information that are not fully represented in final statistical summaries. A spatial field, image, disease map, or set of regional observations may include spatial dependence, local heterogeneity, scale effects, and uncertainty. In statistical practice, however, these complex structures are often reduced to estimated parameters, smoothed surfaces, clusters, risk measures, test results, or prediction errors. Such reductions are useful and sometimes necessary for interpretation, communication, and decision-making, but they may also hide important parts of the original spatial information. This talk discusses information loss in spatial statistics from an entropy-based perspective. Entropy provides a natural way to think about information, uncertainty, and reduction in spatially structured data. The aim is to examine how spatial information changes when data are aggregated, smoothed, classified, or summarized, and to highlight the role of entropy, Bayesian inference, and uncertainty quantification in understanding what is preserved, simplified, or lost during spatial statistical analysis.

Keywords: Spatial statistics, Entropy, Information loss, Spatial dependence, Scale effects, Uncertainty quantification.

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